

Handheld near infrared spectrometer and machine learning methods applied to the monitoring of multiple process stages in industrial sugar production

Bruno Henrique da Silva Melo^a, Rafaella Figueiredo Sales^b, Lourival da Silva Bastos Filho^b, Jorge Souza Povoas da Silva^b, Aluska Gabrielle Carolino de Almeida Sousa^b, Deborah Maria Camará Peixoto^b, Maria Fernanda Pimentel^{b,*}

^a Department of Fundamental Chemistry, Federal University of Pernambuco, Av. Jornalista Aníbal Fernandes, 50.740-560, Cidade Universitária, Recife, Brazil

^b Department of Chemical Engineering, Federal University of Pernambuco, Avenida dos Economistas, 50.740-590, Cidade Universitária, Recife, Brazil

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ABSTRACT

This work aimed to evaluate the performance of a handheld NIR spectrometer in developing calibration models to quantify brix and pol at various stages of an industrial sugar production process. Because of sample variability, collected over two harvesting seasons, NIR measurements were acquired either in transmittance or diffuse reflectance. For modelling purpose, partial least squares (PLS), also combined with variable selection techniques, and support vector machine regression (SVR) were investigated. SVR was applied to handle non-linearities within the data. In general, results illustrated the best performance of SVR, that yielded lower root mean square error of prediction (RMSEP) values for brix and pol for spectra acquisition in transmittance (0.59 and 0.69%w/w) and using diffuse reflectance (1.44 and 2.44%w/w), respectively. Results from models using spectra collected in transmittance were comparable to those reported in other works where benchtop instruments were used, highlighting the cheaper and simpler employment of the portable spectrometer.

1. Introduction

Sugarcane is considered an economically important crop in many countries. It is the main raw material employed in sugar production around the world (Maraphum et al., 2020; Udompetaikul et al., 2021). Key quality parameters such as brix (concentration of total dissolved sugar) and pol (sucrose content) are frequently determined in routine analysis of sugarcane. They are also monitored during multiple stages of sugar production. Traditionally, samples are collected from each step and analyzed in the laboratory using off-line analytical techniques, that require specific equipment and dilution stages for most of the factory samples. This approach, however, delays information about the process, which may cause inefficiency in terms of process monitoring and control (Mehrotra & Siesler, 2003).

Alternative methods that deliver reliable information about the process streams in real-time offer a potential solution to speed up process monitoring, control and decision-making. Soft sensors (or software

sensors), for example, are suitable for this purpose, able to exploit the ever-growing computer capacity and recent developments in data science and machine learning algorithms (Rato et al., 2020). Soft sensors for regression applications consist of calibration models based on on-line measurements, relatively easy to obtain, used for on-line prediction of target variables/parameters that cannot be automatically determined or can only be measured at low frequency and/or high cost (Luttmann et al., 2012; Souza & Araújo, 2016).

There are three major modelling approaches for soft sensor development. They can be model-based (based on phenomenological knowledge), data-driven (based on empirical data of the process) or hybrid, which combines these two approaches. In addition, they can be used based on process variables, for example, temperature, pressure or flow rate (Kadlec et al., 2009; Souza & Araújo, 2016). Regarding application to the sugarcane industry, these three modelling strategies have been employed to develop soft sensors to predict critical parameters of sugar crystallization and sugarcane juice clarification processes

* Corresponding author at: Department of Chemical Engineering, Federal University of Pernambuco, Avenida dos Economistas, 50.740-590, Cidade Universitária, Recife, Brazil.

E-mail address: mfernanda.pimentel@ufpe.br (M.F. Pimentel).

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(Damour et al., 2010; Meng et al., 2019; Meng et al., 2021).

Predictive models for industrial applications based on spectroscopic measurements have also been used as soft sensors. Near infrared (NIR) spectroscopy, for example, is one of the techniques often applied as a Process Analytical Technology (PAT) tool for this purpose (Luttmann et al., 2012; Rato et al., 2020). It is a type of vibrational spectroscopy that covers a wavelength range from 750 to 2500 nm. This region comprises absorption bands mainly corresponding to exciting overtones and combinations of fundamental vibrational modes. It has been increasingly applied to measure product quality and process monitoring using in-situ, in-line, and on-line analytical sensors. The attractiveness of this technique is mainly related to its non-destructive nature, accuracy, short analysis time as well as its feasibility for real-time process monitoring (Pasquini, 2018).

In Brazil, the application of NIR spectroscopy to quantify sugarcane quality parameters used to estimate the payment made to growers is regulated for brix and pol analysis in extracted juice (ABNT NBR 16770: 2019), as well as for brix, pol and prepared sugarcane fiber determinations (ABNT NBR 16771: 2019). In the literature, this technology has been investigated to determine critical parameters of different factory samples, including the sugarcane, the intermediate process streams and the raw sugar. For example, this technique has been applied to determine attributes such as brix, pol and reducing sugars (fructose and glucose) in sugarcane juice (Asprilla & Ramírez-Navas, 2018; Sorol et al., 2010; Valderrama et al., 2007). It has also been used in analyzing molasses samples after dilution to quantify various quality parameters, presenting satisfactory results for fructose, glucose, conductivity ash and dry solids (Simpson & Oxley, 2008). Additionally, on-line analyses of cane, bagasse and raw sugar with NIR spectroscopy have been reported as well (O'Shea et al., 2010). Benchtop instruments were employed in all aforementioned works.

In relation to the cane quality monitoring, classification models based on on-line NIR data from shredded cane have been developed to discriminate between typical and atypical cane samples, in order to identify samples with abnormally low quality (Sexton et al., 2020). Moreover, a number of works have investigated the use of portable instruments in the NIR and visible and shortwave near infrared (Vis/SW-NIR) regions to measure quality parameters such as fiber, brix, pol, commercial cane sugar and solid density in sugarcane stalks (Nawi et al., 2013; Phuphaphud et al., 2020; Sansechan et al., 2018; Taira et al., 2013). These works were mainly focused on in-field applications. In another paper, the Vis/NIR technology has been used in real-time assessment of brix in sugarcane billets moving on an elevator conveyor (Udompetakul et al., 2021). A portable hyperspectrometer in the Vis/NIR region has also been employed to visualize brix and moisture content in sugarcane stalks (Maraphum et al., 2020).

Despite the extensive literature about the use of NIR spectroscopy in the sugarcane industry, few works have described its use to monitor various process streams in the sugar production. For example, Simpson & Naidoo (2010) have reported the application of this technology to the analysis of clear juice, syrup, B, C massecuite and A, B molasses. A NIR benchtop spectrophotometer was used to acquire spectra in the range 800–2500 nm using a 1 mm optical path flow-through cell. Samples were collected during 2008/2009 season from different African sugar factories and sent to the laboratory to be analyzed for brix, pol and sucrose determinations. Except for the clear juice, all the samples were diluted before spectra acquisition. For calibration purpose, partial least squares (PLS) regression models were developed for each type of sample and, in general, satisfactory results were obtained.

More recently, Tange et al., (2015) have described the use of NIR spectroscopy to predict pol and brix of samples collected from four process stages of a Japanese sugar factory during one harvesting season. Vis-NIR transmittance spectra measuring from 400 to 2498 nm were acquired using a bench instrument. Samples were analyzed immediately (without dilution) after milling, evaporation, crystallization and centrifugation, considering the three cycles of crystallization and

centrifugation. PLS and support vector machine regression (SVR) methods were employed to construct local (for each stage) and global regression models. In general, SVR performed better than PLS, which was ascribed to the non-linearities present in the data. In addition, the global SVR models presented satisfactory results for process monitoring. Afterwards, the dataset used in this study was also employed in other works describing the application of different modelling approaches, including alternative non-linear methods and feature selection techniques (Almeida et al., 2018; Ramírez-Morales et al., 2016; Tange et al., 2017).

The present work aimed to evaluate the performance of a handheld NIR spectrometer in monitoring the complete production process of a sugarcane mill. Samples were collected throughout the process (cane reception, milling, treatment system, evaporation, boiling, crystallization, centrifugation and refining) and directly measured with transmittance or diffuse reflectance, depending on the type of sample (juice, concentrated juice, syrup, molasses, massecuite, liquor and magmas). For modelling purpose, PLS (also combined with variable selection tools) and SVR calibration models were developed to assess brix and pol contents in these samples. To the best of the author's knowledge, this constitutes, to date, the first description of the use of a handheld NIR spectrometer for monitoring various process streams in sugar production. Although the handheld NIR instrument has a reduced spectral range and resolution compared to bench instruments, features such as size, simplicity and low cost are its main advantages (Modroño et al., 2017). Finally, the complex nature and large variability of the samples under analysis acquired from different stages of an industrial process, during two harvesting seasons (2019/2020 and 2020/2021), was considered a further challenge to this work.

2. Materials and methods

2.1. Industrial samples

Dataset utilized in the present work were acquired in a sugarcane mill located in the state of Pernambuco – Brazil. In total, 1526 samples were collected from different stages throughout sugar production over two harvesting seasons (727 from 2019/2020 season, and 799 from 2020/2021). The stages of the process, which include various mechanical and chemical operations, are briefly described below.

Initially, in the cane reception area, the sugarcane stalks are analyzed in the laboratory to determine the concentration of total recoverable sugars. This parameter, calculated as a function of brix, pol and fiber content, is the basis of the payment made to growers. After ascertaining its quality, the cane is discharged onto a feeding table, where impurities are removed by washing. It is then chopped, shredded, and sent to the mills (sets of three rolls where cane is pressed for juice extraction). Sugarcane juice obtained in the first mill (primary juice) is mixed with that obtained in the second mill, forming the mixed juice that is sent to the treatment system. In this stage, methods such as filtration, evaporation and decantation are employed to remove impurities and clarify the juice. Subsequently, clarified juice is concentrated in multiple effect evaporators to achieve a brix content of around 65 % w/w. The final product of this process is the syrup; samples collected from these intermediate production streams -called effects- are here called concentrated juice.

Next, the syrup is sent to a three-boiling scheme, comprising three cycles of boiling, crystallization, and centrifugation. By-products generated in this process consist of different types of massecuite (mass of sucrose crystals mixed with syrup or mother liquor obtained from evaporation), molasses (concentrated solution obtained from sugar centrifugation) and magmas (wet sugars obtained from centrifugation that are recycled for massecuite crystallization). This process results in VHP (very high polarization) sugar, with polarization between 99 and 99.49% and light brown color. Finally, in the refining process, the VHP sugar is dissolved in water, forming a liquor that is sent to the treatment

stage. Afterwards, the clarified liquor is sent to a new cycle of boiling, crystallization and centrifugation, producing refined sugar, with a polarization higher than 99.85% and white color. This sugar is then dried and stored prior to its distribution.

The sampling strategy adopted in this work followed the same procedure as the industrial routine analysis for quality control: samples were collected throughout the process at specific stages (Fig. S1 and Table S1) and analyzed in the laboratory to determine their quality parameters. For the development of this work, part of the sample was also analyzed using NIR spectroscopy. After eliminating outliers, a total of 1507 samples collected during both harvesting seasons were investigated, comprising 711 samples from the 2019/2020 season and 796 samples from the 2020/2021 season. The highly heterogeneous compositions of the samples required that the NIR measurements be acquired in transmittance or diffuse reflectance modes using the handheld instrument. The measurement mode selected depended on the physical and chemical properties of the sample. So, less concentrated liquid samples were analyzed using transmittance; and more concentrated samples, which are in general more viscous and darker, were measured using diffuse reflectance. Table 1 presents an overview of these samples, grouped according to their type (a brief description of each sample is given in Table S1).

2.2. Near infrared spectral acquisition and reference methods

The NIR spectral data were acquired directly from untreated samples using a handheld spectrometer (MicroNIR™ 1700, Viavi Solutions, USA), which operates in the wavelength range between 950 and 1650 nm. This instrument has a Linear Variable Filter (LVF) and a 128-pixel InGaAs detector array. Its optical resolution is lower than 1.25% of center wavelength (e.g. the resolution is 12.5 nm at 1000 nm). Each spectrum was obtained as an average of 100 scans with an integration time of 12.3 ms, recording data every 6.2 nm approximately. A transmittance accessory detailed in Paiva et al. (2015) was employed to obtain absorbance spectra of juice samples utilizing a quartz cuvette. In a previous study, different optical path lengths (2, 5 and 10 mm) were investigated to select the one suitable for spectral collection. For these measurements, 100% transmittance was acquired using a clean empty cuvette; and 0% transmittance was obtained by blocking the passage of the light from the source of the accessory (tungsten mini-lamp) to the spectrometer detector.

Spectral data of the other samples (syrup, massecuite, molasses, magmas and liquor), which are more viscous and concentrated, were recorded using diffuse reflectance. A small glass petri dish containing approximately 3 mL of sample was employed for these measurements

Table 1
Samples collected from different process stages of the sugar production and analyzed by NIR spectroscopy.

Process stage	Type of sample	Number of samples	Collected and analyzed samples
Cane reception	Juice	329	Extracted juice
Milling		133	Primary, residual and mixed juices
Treatment system		293	Filtered, evaporated and decanted juice
Multi effect evaporation	Concentrated juice	138	First to fifth effects
Syrup tank (after evaporation)	Syrup	50	—
Crystallisation and boiling	Masseccuite	212	A, B, C and refined masseccuite
Centrifugation	Molasses	169	A, B, C and refined molasses
	Magmas	57	Magmas (2nd and 3rd)
Refining	Liquor	126	Dissolved, mixed, filtered and floated liquor

and the spectrum related to 100% of diffuse reflectance was obtained using a white Teflon board placed close to the MicroNIR window. In relation to the concentrated juice from the multiple effect evaporators, the more viscous samples from the 3rd effect on (brix ranging from 45 to 65 % w/w) were analyzed using diffuse reflectance. Twenty-six samples from the 1st and 2nd effects were analyzed using both transmittance and diffuse reflectance. Both procedures were used in order to investigate which method of spectra acquisition would be more appropriate for these analyses.

Moreover, considering these samples, all of them were analyzed by the reference method in order to quantify brix; and 769 samples were also submitted to pol determination. The former parameter was determined by refractometry, using an automatic digital refractometer (RDA 9000, Acatec, Brazil). This method consists in determining the refractive index of an aqueous solution. Among other factors, the deflection of light is mainly caused by the presence of dissolved solids, enabling their quantification. For sugar solutions, this determination is expressed as brix. Owing to their high viscosity, samples of syrup, massecuite, molasses and magmas were diluted before this analysis, according to the procedure described in Caldas (2012). Depending on the type of sample, two, five, six or eight dilutions were used (Table S2).

To quantify pol, samples were analyzed by polarimetry, using an automatic saccharimeter (Autopol 880, Rudolph Research Analytical, USA). This consists in a measurement of the light polarization, that occurs when a beam of plane polarized light passes through an optically active molecule and deflects its plane of rotation. This property depends, among other variables, on the sample concentration of optically active substances; in a sugarcane mill, this is mainly due to D-sucrose. This determination was performed after a clarification step using lead acetate for most of the samples (to remove colloids and impurities that give the sample color) and reading in a saccharimeter at wavelengths 587 and 589.4 nm (Caldas, 2012).

2.3. Data treatment and modelling

As NIR data were acquired for both transmittance and diffuse reflectance, a single model for all the samples could not be considered under the conditions of this work. Our goal, despite this, was to obtain regression models comprising different samples that could be analyzed in the same way. This strategy aimed to simplify the application and maintenance of the calibration models in process monitoring, making it necessary to define first which was the best approach to analyze samples with intermediate values of brix (those concentrated juice samples from the 1st and 2nd effects measured with both spectra acquisition procedures). Only brix was evaluated in this preliminary study, since these samples had not been quantified for pol analysis in this particular industrial routine. The other models described in this work were developed for both brix and pol.

In this initial study, two PLS models were developed for each spectral acquisition procedure: one including twenty-six concentrated juice samples available from the 1st and 2nd effects of the evaporation process stage (which were analyzed in transmittance and diffuse reflectance) and the other one without these samples. The distribution of these samples in the calibration and validation sets were similar for both the transmittance and diffuse reflectance models. To construct all these PLS models, spectral data were preprocessed with Savitzky Golay first derivative (adopting a second order polynomial function and a 5-point window) and mean-centered.

After defining the spectral acquisition procedure for these samples and, consequently, the concentration range of each model for brix determination, we built PLS models to quantify brix and pol by using different preprocessing techniques. The following strategies were evaluated: standard normal variate (SNV), multiplicative scatter correction (MSC) and Savitzky-Golay first and second derivative (adopting a second order polynomial function and a 5- or 7-point window). After preprocessing, the data were mean-centered for modelling purpose. In

order to evaluate the models constructed based on the raw data, we also built PLS models without applying the other types of pre-processing (except for mean-centering). In order to build and validate all the PLS models, the samples set were split, using the Onion method (Olsson et al., 2004), into calibration and validation sets in a proportion of about 70% and 30%, respectively.

The preprocessing technique that led to the best predictive performance for the PLS models using the full spectra was then used in the application of variable selection techniques: variable importance in projection (VIP) (Andersen & Bro, 2010) and sparse PLS (sPLS) method (Calvini et al., 2015). In addition, SVR models were also developed using the selected preprocessing tool. Although PLS is one of the most common methods for analyzing chemical data, being relatively easy to interpret, it is not able to cope with highly non-linear data. In these cases, methods such as SVR, which are capable of accurate accounting for non-linearities and generalizing, can be used (Tange et al., 2015; Tange et al., 2017). For details about this machine learning algorithm, the reader is referred to the references (Brereton & Lloyd, 2010; Smola & Scholkopf, 2004).

Although the PLS method has been properly used to quantify brix and pol in specific types of samples in the sugar cane industry such as sugarcane juice (Sorol et al., 2010; Valderrama et al., 2007), stalk (Nawi et al., 2013; Taira et al., 2013; Maraphum et al., 2020) and billet (Udompettaikul et al., 2021), it is important to highlight that our work aimed to build more global models, including samples from different process stages. Among other factors, this sample variability may lead to non-linearities in the data, justifying the application of the SVR, as discussed in (Tange et al., 2015). Our SVR models were developed using the radial basis function (RBF) kernel type. For each model, a two-step grid search was used to find a combination of the parameters (cost - C, epsilon - ϵ and gamma - γ) that provided a minimum value of root mean square error of cross-validation (RMSECV). In the first step, the grid search was performed using default value ranges for these parameters. In the further step, these ranges were adjusted based on the results obtained in the first step (surface charts of RMSECV generated for each two parameters). To prevent overfitting, these models were properly validated, using the same calibration and validation sets as employed in the corresponding PLS models. In addition, the figures of merit of the models and their number of support vectors (SVs) were evaluated in order to select between the SVR models generated in the first and second steps.

Outliers were detected using residual vs. leverage plots (95% confidence level) generated by PLS modelling and through the spectral analysis. They were identified for each dataset / model, and in general were detected as random, spot observations. After their exclusion, the models were rebuilt without these samples (see Table S3 for additional details about number of samples and outliers for each dataset). The outliers detected and excluded in this approach were also removed from the datasets prior to the development of the SVR models, because of the limitations in interpreting and exploring SVR models (Tange et al., 2015).

The number of latent variables (LVs) in the PLS models was determined using venetian blinds cross-validation, also used to select the optimal SVR parameter values. As mentioned above, the performance of the models was assessed using validation sets and evaluating figures of merit such as the root mean square error of prediction (RMSEP) and bias.

Data treatment was carried out on Matlab (MATLAB® R2010a 7.10.0.499, MathWorks, USA) and all regression models were developed using PLS Toolbox (Eigenvector Research Inc., USA).

3. Results and discussion

3.1. Selection of the optical path length for transmittance measurements

Mean spectra of juice samples acquired using cuvettes with optical paths of 2, 5 and 10 mm (Fig. S2) were evaluated in order to select the

best ones for transmittance measurements. The spectra acquired with 5 and 10 mm path length cuvettes showed saturated absorption signals in the region between 1400 and 1650 nm, the relevant wavelengths for quantifying both brix and pol (Almeida et al., 2018; Ramírez-Morales et al., 2016). The absorption around 1400 and 1550 nm were assigned, respectively, to the first overtone of O — H bonds from water and sucrose (Workman & Weyer, 2012). Saturated signals in this region were then attributed to the high water content of juice. The spectrum acquired using a 2 mm path length cuvette presented a broad but well-defined band in this region; thus, it was used to analyze juice and concentrated juice samples.

3.2. Definition of the spectral acquisition procedure for concentrated juice samples from the 1st and 2nd effects of the multiple effect evaporator stage

One of the main steps of sugar production is carried out in multiple effect evaporator systems, where process streams become more concentrated and viscous with the progress of the process. As mentioned above, samples from the 3rd effect forward were analyzed using diffuse reflectance, while most of the samples from the 1st and 2nd effects were measured using both procedures of spectra acquisition to evaluate in which model to quantify brix these samples should be included. In addition, part of the samples from the 2nd effect (with brix content higher than 45 % w/w) were only analyzed in diffuse reflectance due to their high viscosity. In general, these effects present overlapped concentrations. Thus, it was not possible to specify a single acquisition procedure to analyze samples from the 2nd effect, as these samples contained a wide variation in brix, limiting establishment of the ranges of the models. This drawback, nevertheless, was not considered a critical issue as it was not vital for process monitoring.

The results of the PLS models developed with and without concentrated juice samples from the 1st and 2nd effects (with brix lower than 45 % w/w) are presented in Table 2. For diffuse reflectance, the RMSEP values obtained for each model were statistically different, based on an F-test, at a confidence level of 95% (see Table S4 for detailed information of the tests). On the other hand, for transmittance, these values were equivalent, based on the same statistical test. In general, low bias values were observed for these models, implying that there is no evidence of systematic errors (based on a *t*-test, at a confidence level of 95%, as recommended by (ASTM E1655-17:2017)).

Looking at PLS quantitative results of brix (Fig. 1) it is possible to see that the model for transmittance data set (Fig. 1a) closely fits the concentrated juice samples from the 1st and 2nd effects, despite the slight separation between juice and concentrated juice samples in the range around 25–30 % w/w. In contrast, an inferior fit of these samples was observed for the diffuse reflectance data set model (Fig. 1b). Based on these findings, samples from the 1st and 2nd effects, with brix lower than 45 % w/w, were included in the model developed using transmittance measurements.

3.3. Preprocessing procedures and modelling

From the preprocessing techniques employed in the development of PLS models using the full spectral range, Savitzky-Golay first derivative showed the best performance. In general, the models that used this strategy were constructed with one more latent variable (LV) compared to SNV and MSC. Nevertheless, they yielded lower RMSEP values, according to a randomization *t*-test (van der Voet, 1994) at 95% confidence. In addition, this preprocessing was more efficient in correcting scattering effects present in the spectra, especially for those obtained using diffuse reflectance. It could also distinguish between different types of samples, which presented superimposed profiles. This may have led the need for one more LV.

By comparing the Savitzky-Golay first and second derivatives, in general, the corresponding RMSEP values were not statistically different (randomization *t*-test, 95% confidence), but the second derivative

Table 2

Results for calibration and prediction of the PLS models with and without the concentrated juice samples from the 1st and 2nd effects from the multiple effect evaporator stage (with brix lower than 45 % w/w).

	1st and 2nd effects	Range (% w/w)	LVs	RMSEC (% w/w)	RMSECV (% w/w)	R^2_{CV}	RMSEP (% w/w)	R^2_P	Bias (% w/w)
Transmittance	With	1.02 – 45	4	0.71	0.72	0.98	0.74	0.99	−0.001
	Without	1.02 – 25.5	4	0.64	0.65	0.98	0.72	0.98	−0.036
Diffuse Reflectance	With	30.6 – 95.3	4	2.62	2.67	0.97	2.65	0.97	−0.06
	Without	45.3 – 95.3	4	2.25	2.33	0.97	2.36	0.97	−0.03

LVs: number of latent variables.

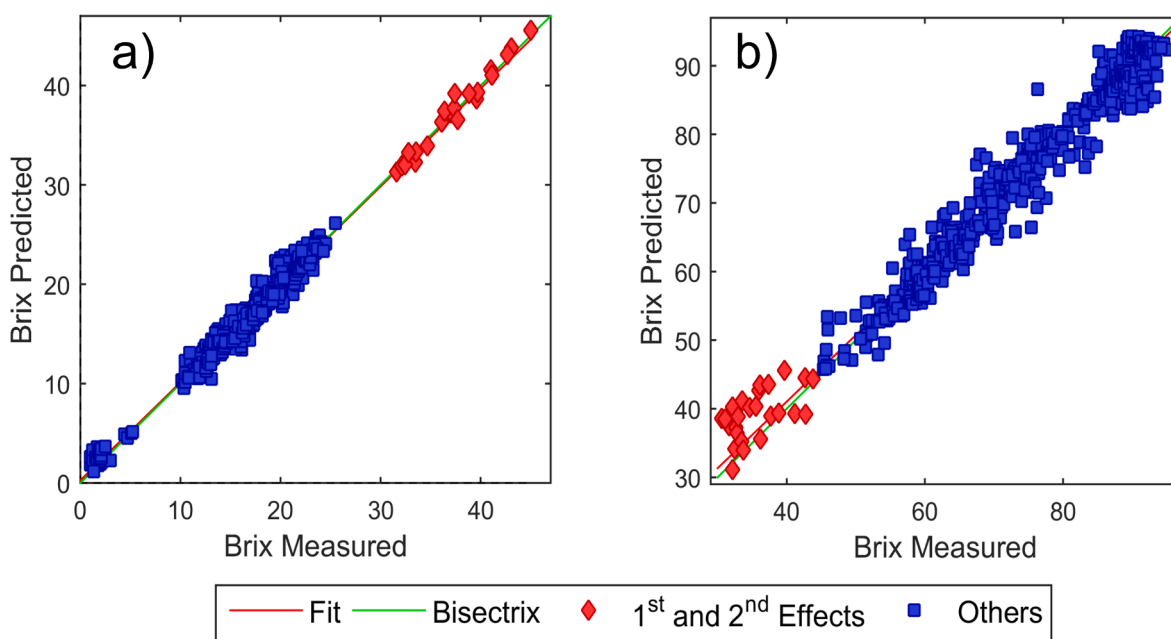


Fig 1. Predicted versus reference plots for the models with concentrated juice samples from the 1st and 2nd effects from the multiple effect evaporator stage (with brix lower than 45 % w/w) for transmittance (a) and diffuse reflectance (b).

required more LVs. In addition, similar performance in terms of prediction power and number of LVs were obtained regardless of the use of a 5- or 7-point window. Finally, the models built with Savitzky-Golay first derivative presented lower RMSEP values (randomization *t*-test, 95% confidence) compared to those obtained without preprocessing (the only exception was for the pol model for diffuse reflectance data, that presented similar results). Based on these findings, the Savitzky-Golay first derivative with a window of 5 points was selected.

Fig. 2a shows the raw spectra of 781 juice and concentrated juice samples (from the 1st and 2nd effects from the multiple effect evaporator stage) analyzed using transmittance. A moderate scattering effect is present as a consequence of insoluble solids in suspension. Because of high water content, these spectra are dominated by the water absorption bands. The strong and broad band around 1450 nm can be ascribed to the first overtone of O — H bonds from water and sucrose and C — H combination bands for sugars (Mehrotra & Siesler, 2003). It is also possible to see another characteristic water absorption band around 979 nm (second overtone of O — H bonds) (Workman & Weyer, 2012). The mean spectra of each type of sample corrected with SNV are displayed in Fig. 2b. In this figure, one can observe that the spectral profiles of these samples are similar, with a slight difference in the region between 1400 and 1650 nm, that can be seen after application of first derivative transformation (Fig. 2c). As mentioned above, this region comprises the sucrose absorption bands overlapping with the broad water band.

Also depicted in Fig. 2 are the raw (d), mean (e) and preprocessed (f) spectra of the 726 samples analyzed using diffuse reflectance. The variations reflected in the spectral profiles occur because these samples

range from concentrated juice (viscous liquids) to magmas (wet sugars, predominantly solids). In order to visualize and compare these overlapped profiles, Fig. 2d shows the mean spectra of each type of sample as well as the mean spectrum of refined sugar, corrected with SNV. It is possible to see that with the evolution of the production process, these spectral profiles resemble the sugar spectrum. This similarity is emphasized for magmas, that have a high sucrose concentration. Fig. 2f shows the preprocessed data using Savitzky-Golay first derivative, that was applied to correct baseline shift. These scattering effects are common to diffuse reflectance measurements and reflected to a greater extent in the spectra of molasses, massecuite and magmas (Fig. 2d). This is probably associated to the heterogeneity of these sample sets, as they were different in composition, viscosity, temperature and crystal size. After preprocessing, most of the baseline shift was corrected and the main sources of variation that remained were observable in the regions near 1000, 1200 and 1400 nm, assigned to water and sucrose absorption bands (Osborne, B.G.; Fearn, T.; Hindle, P.H., 1993).

Owing to the best performance of the first derivative, a comparison between the PLS and the SVR approaches was made. In relation to the use of variable selection techniques, the PLS models for the transmittance data obtained applying the VIP method led to similar results compared with the corresponding PLS models using the full spectra. For the diffuse reflectance data, the VIP method produced PLS models with inferior results. Regarding the sPLS models, intermediate RMSEP values were obtained using this method. Only two of the sPLS models presented RMSEP statistically lower than those obtained with PLS (full spectra), according to a randomization *t*-test, at a confidence level of 95%.

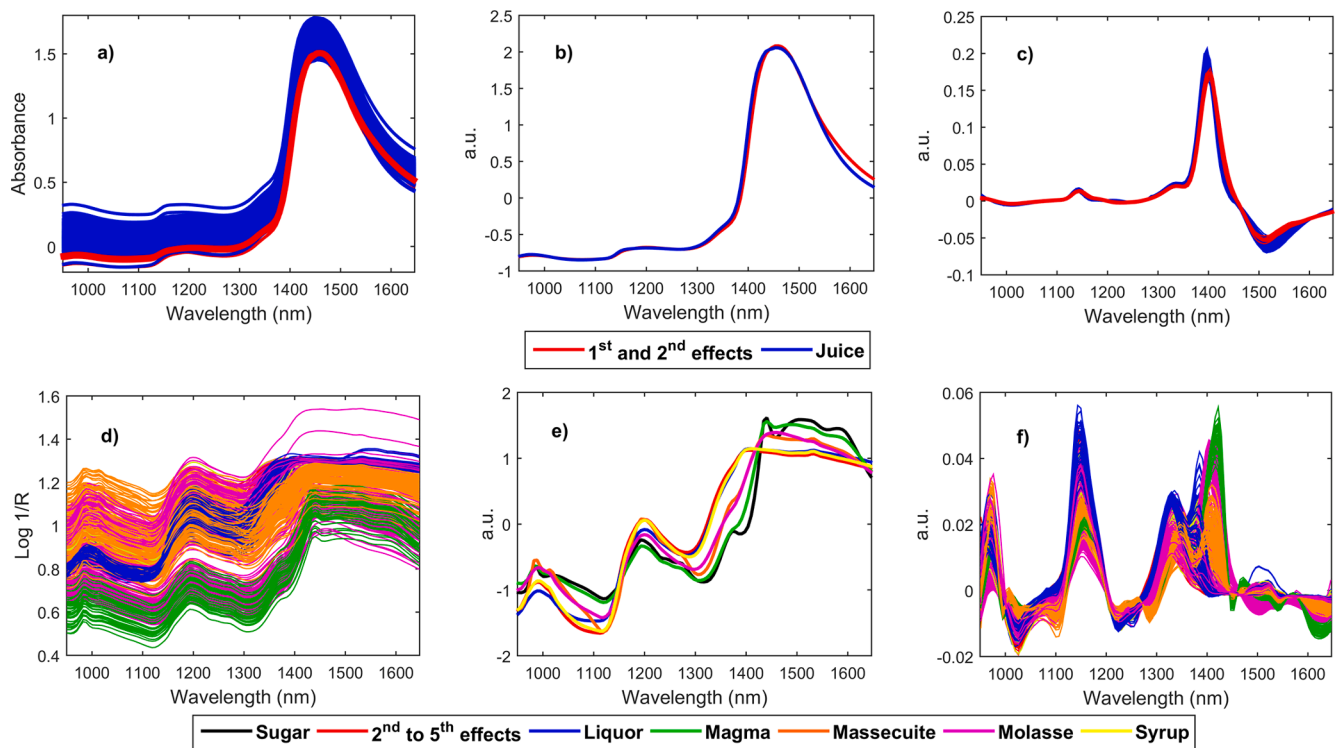


Fig 2. Raw (a, d), mean (b, e) and preprocessed with Savitzky-Golay first derivative (c, f) spectra acquired using transmittance and diffuse reflectance, respectively, and coloured according to the type of sample.

Nevertheless, when compared with SVR, these models presented inferior results, based on the same statistic test. Therefore, as the variable selection techniques employed in the current study did not show an overall improvement in the PLS results, for brevity purpose, we have decided to present a comprehensive comparison between the PLS and SVR models using the full spectral range.

After removal of the outliers, the PLS and SVR models to quantify brix and pol for transmittance spectra were built, respectively, with 547 and 324 calibration samples and using 234 and 138 samples in the external validation. The models to determine brix and pol for diffuse reflectance spectra were elaborated, respectively, with 508 and 215 samples and validated using 218 and 92 independent prediction samples. The figures of merit of these models are outlined in Table 3, considering the best SVR models selected after the two-step optimization. In general, they presented no significant bias, according to a *t*-test, at a confidence level of 95% (ASTM E1655-17:2017).

Considering the PLS models for transmittance measurements (Table 3), RMSEP values presented for brix (0.74 % w/w) and pol (0.77 % w/w) were comparable to those obtained in other works. For example, to estimate brix in sugarcane juice using transmittance measurements acquired with a Vis-NIR benchtop spectrometer, Sorol et al. (2010) obtained RMSEP values of 0.52 % w/w for the full spectral range

(400 to 2498 nm) and 0.28 % w/w using a genetic algorithm for variable selection. In these models, brix varied from 11.76 to 23.15 % w/w, with this range being considerably narrower compared to the applied in this work. In addition, values of 0.28 % w/w for brix and 0.42 % w/w for pol were obtained by Valderrama et al., (2007). To build these models, a benchtop NIR spectrometer operating in the spectral range of 1100–2500 nm was used to analyze sugarcane juice using transmittance. The working concentration range used for brix (8–26 % w/w) was also smaller than the one employed in the current study, whereas for pol (4–23 % w/w) it was similar. In contrast, the PLS models using diffuse reflectance spectra presented higher RMSEP values (Table 3) in comparison to those reported in the literature (Tange et al., 2015).

The predicted versus reference plots of these PLS models are graphically displayed in Fig. 3 (a, d, g, j) together with the corresponding plots of Variable Importance in Projection (VIP) scores (Fig. 3c, f, i, l). A variable with a VIP score close to or greater than one can be considered important in the projection used in the PLS model. Although the VIP scores are frequently used for variable selection, these plots were generated from the PLS models developed using the full spectra with the aim of evaluating and interpreting these models. Looking at Fig. 3 (c and f), we can see that the most relevant variables for transmittance models are in the regions around 1400 and 1500 nm (first

Table 3

Results for calibration and prediction of PLS and SVR models to quantify brix and pol obtained for samples analysed in transmittance or diffuse reflectance modes.

		QP	Range (% w/w)	LVs or SVs	RMSEC (% w/w)	RMSECV (% w/w)	R ² _{cv}	RMSEP (% w/w)	R ² _p	Bias
PLS	Transmittance	Brix	1.0–45	4	0.71	0.72	0.98	0.74	0.99	−0.001
		Pol	1.1–20.6	4	0.74	0.76	0.95	0.77	0.95	0.01
	Diffuse reflectance	Brix	45.3–95.3	4	2.25	2.33	0.97	2.36	0.97	−0.08
		Pol	36.2–90.0	5	2.86	2.99	0.97	2.98	0.97	0.12
SVR	Transmittance	Brix	1.0–45	465	0.59	0.61	0.99	0.59	0.99	0.015
		Pol	1.1–20.6	184	0.62	0.69	0.96	0.69	0.96	0.05
	Diffuse reflectance	Brix	45.3–95.3	478	1.35	1.64	0.98	1.44	0.99	−0.08
		Pol	36.2–90.0	96	2.20	2.43	0.98	2.44	0.98	−0.13

QP: quality parameter; LVs: latent variables for PLS; SVs: support vectors for SVR.

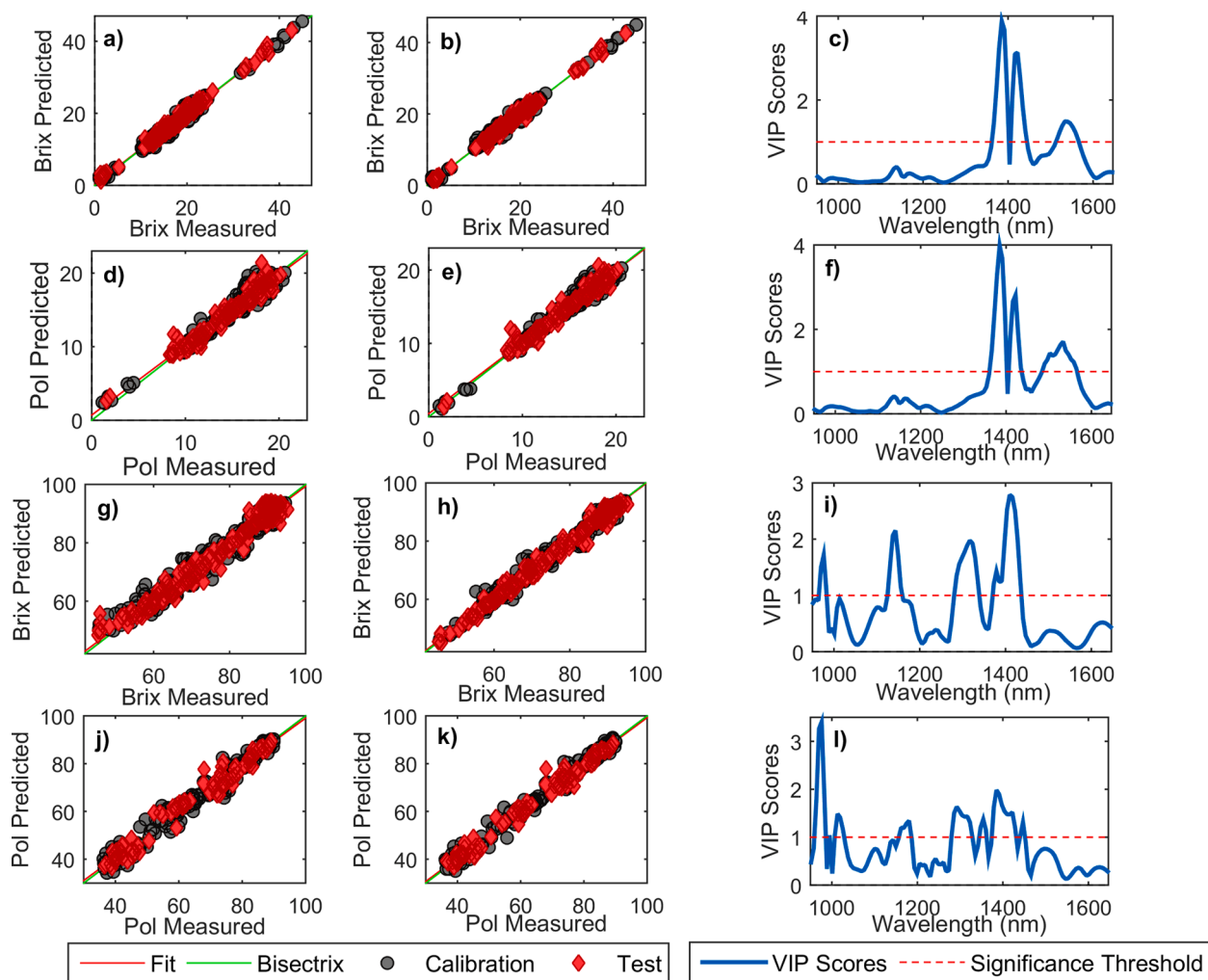


Fig 3. PLS (a, d, g, j) and SVR (b, e, h, k) quantitative results and VIP Scores plots for the PLS models (c, f, i, l) for: brix models for transmittance (a, b, c); pol models for transmittance (d, e, f); brix models for diffuse reflectance (g, h, i); and pol models for diffuse reflectance (j, k, l).

overtone of O—H from water and sucrose, respectively). The equivalence between these plots may be related to the apparent purity of the samples included in these models, which was higher than 80%. This is the ratio of pol to brix expressed in percentage, implying that these samples have similar concentrations of these parameters. For the diffuse reflectance models (Fig. 3i and l), besides 1400 nm, other regions attributed to the sucrose molecule were also noticeable: 950 nm (second overtone of O—H bonds), 1150 nm (second overtone of C—H stretch) and 1300 nm (combinations of C—H group). The apparent purity of the samples presented more variations, ranging from 42% (for C molasses) to 99.9% (for refined massecuite), leading to the differences observed in the VIP scores. In general, these regions agreed with those selected in other works, using different variable selection techniques, for modelling of brix and pol in sugar production (Almeida et al., 2018; Ramírez-Morales et al., 2016; Sorol et al., 2010).

Regarding the SVR approach, all the SVR models presented lower RMSEP values (Table 3) compared to the corresponding PLS models, according to a randomization *t*-test (van der Voet, 1994), at a confidence level of 95% (see Table S5 for detailed information of the tests). This corroborates the presence of non-linearities in the dataset (as the non-linear trend observable in Fig. 3g), which led to a reduction in the PLS performance in comparison to SVR. These findings agree with previous results that modelled the multiple process steps of sugar production (Tange et al., 2015; Tange et al., 2017). In general, the sources of non-linearities are varied and difficult to detect. In the NIR data used in

this work, acquired from samples collected from different stages of the production process, non-linear sources may have been originated from some of the following (Bertran et al., 1999; Miller et al., 1993; Nawi et al., 2013; Sexton et al., 2018; Tange et al., 2015; Tange et al., 2017): high concentrations of brix and pol; heterogeneity of the chemical composition of samples; variations in temperature, viscosity, particle size of sucrose crystals and pH.

A general look at Fig. 3 and Table 3 indicates that the dataset most benefited from the application of SVR were the samples of the brix model that used diffuse reflectance. A reduction of about 40% in the RMSEP value was obtained by applying SVR instead of PLS to these diffuse reflectance measurements used to model brix. This behavior is expected, as this dataset, varying from concentrated juice to magmas, is affected more highly by the aforementioned non-linear effects. Regarding the brix model for transmittance and the pol model for diffuse reflectance data sets, their RMSEP values reduced about 20% with the application of the SVR method, while for the pol model for transmittance measurements, this reduction was, approximately 10%. Indeed, samples analyzed in transmittance (mostly juice) presented lower concentrations of the analyte and were measured in general at lower temperatures, suggesting that they were less influenced by deviations from linearity.

Finally, the results of the SVR models developed in this work, using a handheld instrument working in the spectral range from 950 to 1650 nm, were compared to previous results reported in the literature (Almeida et al., 2018; Ramírez-Morales et al., 2016; Tange et al., 2015).

All these works used the same dataset, consisting of NIR transmittance measurements acquired from four process steps of sugar production using a benchtop spectrometer (400–2498 nm). Comprising these steps, global models were developed covering a concentration range of 15–95 % w/w for brix and 13–78 % w/w for pol. In this context, the brix model obtained in the current study for transmittance data, with brix ranging between 1.0 % w/w and 45 % w/w, presented a comparable RMSEP value (0.59 % w/w) to that reported by (Tange et al., 2015) (RMSEP = 0.59 % w/w) and greater than that reported by (Ramírez-Morales et al., 2016) (RMSEP = 0.305 % w/w), although the ranges are not the same.

For a higher brix content (45.3 % w/w to 95.3 % w/w), the diffuse reflectance spectral acquisition was used for the industrial samples. In such case, the RMSEP value obtained in this work (1.44 % w/w) was greater than that presented in (Tange et al., 2015) (RMSEP = 0.59 % w/w), but similar to the RMSEP value obtained by (Almeida et al., 2018) (RMSEP = 1.0 % w/w). As mentioned above, Almeida et al. (2018) used the same dataset as described in (Tange et al., 2015), but applied iSPA-Kernel-PLS for modelling purpose instead of SVR.

For pol in less concentrated samples (with pol varying between 1.1 and 20.6 % w/w), spectral data were obtained in transmittance using the portable instrument. In this case, the RMSEP value obtained (0.69 % w/w) was also similar to those reported in (Tange et al., 2015) (RMSEP = 0.64 % w/w) and (Ramírez-Morales et al., 2016) (RMSEP = 0.485 % w/w). It was lower, however, compared to that obtained in (Almeida et al., 2018) (RMSEP = 1.1 % w/w), using i-SPA-Kernel-PLS.

For a higher pol range (36.2–90.0 % w/w), the SVR model built in this work presented a higher RMSEP value than what has been reported in previous studies (Ramírez-Morales et al., 2016; Tange et al., 2015). However, the relative RMSEP (estimated with relation to the average of the concentration range) for this model was about 3.9%, which is considered a very satisfactory performance regarding the benefit-cost ratio of the ultra-compact spectrometer used.

Despite this inferior prediction performance achieved for pol, for more concentrated samples analyzed in diffuse reflectance, the outcomes obtained by the current study are very promising for the target of process monitoring. In fact, the strategy still needs to be evaluated in terms of fit-to-purpose, not just based on the figures of merit of the models compared to those obtained using benchtop instruments. As discussed by Pasquini (2018), this would be a more appropriate criterion in the decision to use a prediction model constructed based on measurements collected with miniaturized spectrometers. It is important to highlight the heterogeneity and complexity of the industrial samples included in this work, which were acquired over two harvesting seasons, from various stages of the sugar production process, encompassing all sample variations discussed above.

4. Conclusion

The potential of a handheld NIR spectrometer was evaluated to determine brix and pol at different stages of the sugar production process (from cane reception to sugar refining). A preliminary study was accomplished to define the best spectral acquisition procedure for analyzing different industrial samples. Because of the great differences of these samples, transmittance and diffuse reflectance were used.

Two regression methods were investigated for modelling purpose. In a first step, PLS models were developed using different spectral pre-processing techniques. The best predictive performance was achieved using Savitzky-Golay first derivative, that was also employed to construct PLS models applying variable selection approaches and SVR models. Due to its ability to deal with non-linearities in the data, the SVR approach outperformed the PLS. Using this method, lower RMSEP values were obtained for brix and pol for transmittance (0.59 % w/w and 0.69 % w/w) and diffuse reflectance (1.44 % w/w and 2.44 % w/w), respectively. The dataset that benefited most from the SVR application contained concentrated and viscous samples including those in the brix

model, analyzed with diffuse reflectance. As discussed above, these samples may have been greatly affected by non-linear effects.

In particular, for transmittance data set, the predictive performance of the SVR models were comparable to those reported in previous studies using benchtop spectrometers. In contrast, the pol model including samples analyzed in diffuse reflectance presented a higher RMSEP value in comparison to other works. Nevertheless, this model's relative RMSEP was still considered satisfactory regarding the benefit-cost ratio of using an ultra-compact instrument, given the complexity of the industrial samples analyzed in this work.

Finally, the findings of this manuscript demonstrated that a handheld NIR spectrometer can be used to develop soft-sensors to monitor important quality parameters in the sugar industry. For each parameter, two regression models only were required to cover all the samples included in this study. This makes the handheld instrument a simpler and cheaper means for process monitoring, which can be adapted to on-line determinations, reducing the need for labor-intensive and off-line reference analysis and, therefore, speeding up process monitoring, control and decision-making.

Regarding the application of the built models for on-line predictions, they would probably require the use of a model adaptation strategy. This could be done, for example, by applying calibration transfer or expanding the calibration set to account for variations caused by the on-line measurements in order to improve model robustness. In this sense, the most critical points for this implementation would be the variations in environmental and sample conditions (e.g. differences in temperature and other physical effects reflected in the spectral data acquired on-line).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodchem.2021.130919>.

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